Ram Green Moduli and Conditional All-Start Limits for Tree-Weighted Planar Walks

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M377

October 8, 2026

Abstract

We analyze the pinned linear-clock manuscript for spanning-tree-weighted planar maps, retaining its all-edge half-edge walk, degree mass, centered Green operator, and half-energy continuum convention. We reconstruct the source-specific dependencies from exact incidence sewing through primal–dual conductivity calibration, finite-map Green estimates, and uniform resolvents. Keeping the source's dyadic cut scales instead of interpolating each scale gives a square-root bound in L^2 of the map law for the small-source modulus at each deterministic mass, with a quadratic tail bound. A summable scale weight gives one map-dependent coefficient, bounded in L^2 and uniformly integrable in L^1, for a simultaneous square-root modulus with any fixed logarithmic power greater than one half. These local refinements use the source's forest identity and one-scale moment estimate, not a new geometric input. Separately, we prove that the source's uniform resolvent and uniform exit conclusions imply a canonical continuous-path law from every limiting point, convergence uniform over discrete starting vertices, and transfer of arbitrary weakly convergent initial laws. This is conditional on stronger inputs than stationary path convergence. Actual private Ram passages clarify mass transport, gauge fixing, and the distinction between strong and uniform operator control; the analytical tools remain classical. Exact finite checks do not certify the source's global quantum-map claims, historical novelty, or a computational speedup. Contribution assessment: C3/5, the quantified near-endpoint Green refinement and arbitrary-weak-start transfer notably strengthen local source conclusions while retaining their essential imported inputs.

1 Scope and Mathematical Responsibility

The assigned source is *A Linear Clock for Random Walk on Tree-Weighted Planar Maps*, October 5, 2026 [1], at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The actual main file, history, all numbered sections from introduction through dynamics, and bibliography were read in full. Selected mathematical passages of its metric companion [2] were also read, with the narrower scope recorded in Appendix A. Reading all of this manuscript does not certify its entire dependency chain.

The report distinguishes three kinds of statement. Equations about a fixed finite graph are proved directly. The new modulus estimates in Section 6 are implications of the source's finite-map one-scale estimate, whose proof is reconstructed here. The path extension in Section 8

is a deterministic implication of uniform resolvents, small jumps, and a uniform exit estimate, applied only conditionally to the source's random-map construction. Its geometric and field-locality dependencies remain imported. No empirical test promotes an imported assertion to a proved global theorem.

The private sources are used at their actual formulas and proofs, not because of their titles. Volume 21's physical-metric chapter [3] transports the Hilbert norm with coefficients; its spectral chapter [4] distinguishes strong from norm resolvents and specifies operator domains. The graph Green report [5], equations `eq:gauge`, `eq:gaugeinv`, and `eq:transport`, supplies a concrete compatible-load and gauge calculation. Their relevance is $L4/5$; this is separate from the realized contribution score $C3/5$. These classical principles are not independent evidence for Liouville universality.

The frozen M375 report [6] is a comparison, not a template for new claims. Its conductance perturbation and L^1 initial-density extensions are inherited and receive no contribution credit here. The present source has a different parameter, model, and analytical route: $\gamma = \sqrt{2}$, spanning-tree weighting, and a finite-map supremum Green bound, rather than the FK–Ising corner-space Hilbert–Schmidt inverse input at $\gamma = \sqrt{3}$.

2 State, Mass, Energy, and Clock

2.1 The exact discrete model

For each integer $n \geq 1$, sample uniformly a rooted spherical planar map with n edges together with a chosen spanning tree. Equivalence preserves the orientation and root. Consequently the marginal map probability is proportional to its number of spanning trees and, given the map, its selected tree is uniform. Loops and parallel edges are permitted. Every walk and graph distance below uses *all* map edges, not just the selected tree. Write $\mathcal{M}_n$ for the full realized map, including its graph and root. Conditioning on $\mathcal{M}_n$ below means a finite uniform average over its spanning trees, not conditioning only on its degree mass matrix M_n.

Let d_v be the degree, with loops counted twice, and put

$$\mu_n(v) = \frac{d_v}{2n}, \quad M_n = \operatorname{diag}(\mu_n(v)), \quad H_n = D_n^T D_n, \quad \mathfrak{e}_n(u) = u^T H_n u. \tag{2.1}$$

Here D_n has one oriented incidence row for each original edge; a loop has a zero row. All finite function spaces are real, and all finite form and operator domains are the whole vertex space. In the one-vertex loop case, $H_n = 0$ but $\mu_n = 1$.

At each rate-one Poisson attempt choose one of the d_v half-edges uniformly. Write P_n for its step matrix, $L_n = P_n - I$, and

$$A_n = -nL_n = \tfrac{1}{2} M_n^{-1} H_n, \qquad \langle u, A_n v \rangle_{\mu_n} = \tfrac{1}{2} u^T H_n v. \tag{2.2}$$

For distinct v, w, $P_n(v, w)$ is the number of edges between them divided by d_v; its diagonal includes twice the loop count divided by d_v. Detailed balance follows because each oriented nonloop incidence has stationary attempt weight $1/(2n)$. Equation (2.2) also follows directly from $H_n = -\operatorname{diag}(d_v) L_n$. It is an exact identity, not a time-scale asymptotic. Invisible loop attempts do not require a further holding-time correction.

The symbol A_n here denotes the positive walk operator; the random Green constants below are denoted $a_{n,p}$ to avoid the source's reuse of A_n for a random bound. For $\mu_n(f) = 0$, define

$$H_n \mathsf{G}_n f = M_n f, \qquad \mu_n(\mathsf{G}_n f) = 0. \tag{2.3}$$

Thus G_n is the inverse of $M_n^{-1}H_n$ on mean-zero functions, whereas the inverse of A_n there is $2\mathsf{G}_n$. Forgetting this factor would change the clock.

2.2 Continuum spaces and exact quantifiers

The limiting surface is the ordinary area-one $\sqrt{2}$ quantum sphere with distinguished points forgotten. It carries quantum probability measure μ_h, intrinsic metric D_h in a fixed deterministic normalization, and a conformal round-sphere coordinate. The source assumes the closure of

$$\mathcal{E}_h(u,v) = \tfrac{1}{2}\int_{S^2} \nabla u \cdot \nabla v \,\mathrm{d}A, \qquad u, v \in C^\infty(S^2), \tag{2.4}$$

in $L^2(\mu_h)$ is a regular conservative Dirichlet form, with quasi-continuous representatives and continuous associated paths. Call its domain $\mathcal{F}_h$. Its positive generator has domain

$$D(A) = \{u \in \mathcal{F}_h : \exists b \in L^2(\mu_h)\ \forall v \in \mathcal{F}_h,\ \mathcal{E}_h(u,v) = \langle b, v\rangle_{\mu_h}\}, \quad Au = b. \tag{2.5}$$

Smooth functions are a form core; they are not silently assumed to be in $D(A)$. A formal expression involving $e^{-\sqrt{2}h}\Delta$ does not replace (2.5). The metric D_h is not the Dirichlet-form intrinsic pseudometric.

For a finite real random variable Y, let $m(Y)$ be the unique zero of $r \mapsto \mathbb{E}\arctan(Y - r)$. Strict decrease, continuity, and limits $\pm\pi/2$ give existence and uniqueness, without moment assumptions. The source uses the deterministic spatial scale

$$s_n = \exp\{m(\log \operatorname{diam}_{D_h} S) - m(\log(1 \vee \operatorname{diam}_{d_n} V_n))\}.$$

The metric companion supplies $s_n \to 0$ and GHP convergence at this scale. This unconditional diameter calibration is not electrical calibration; multiplying the graph generator would leave GHP unchanged.

For every fixed $T \in (0,\infty)$, the source asserts convergence in law of $(V_n, s_n d_n, \mu_n, Q_n^T)$ to (S, D_h, μ_h, Q_h^T), where Q_n^T is the *conditional* law of X_{nt}^n from μ_n given the decorated map, and Q_h^T is the conditional stationary law of the half-energy LBM. The metric on these quadruples takes the infimum over common isometric embeddings of the maximum of Hausdorff distance, Prokhorov distance of masses, and Prokhorov distance of the path laws on Skorokhod $D([0,T])$ with J_1 topology. The conditional law is part of the random state, not merely its annealed average. Fixed finite products of conditional walk laws also converge. Neither a number of walks tending to infinity nor uniformity over time horizons is asserted.

3 Physical Ram Representation

Let μ be any strictly positive probability vector on a connected finite graph, $M = \operatorname{diag}\mu$, $H = D^T D$, and $\Pi = \mathbf{1}\mu^T$. The neutral *load* condition is $\mathbf{1}^T q = 0$; the chosen *potential* gauge is $\mu^T u = 0$. These are different subspaces. The actual private calculation in [5] gives the symmetric core kernel

$$K_\mu = (I - \mathbf{1}\mu^T)H^+(I - \mu\mathbf{1}^T), \qquad HK_\mu = I - \mu\mathbf{1}^T, \qquad K_\mu\mu = 0. \tag{3.1}$$

Here H^+ is the counting-inner-product Moore–Penrose inverse. For centered f, $\mathsf{G}_\mu f = K_\mu M f$; extending it by zero on constants gives $\mathsf{G}_\mu = K_\mu M$ on all functions. Therefore

$$(M^{-1}H)\mathsf{G}_\mu = I - \Pi, \quad \mathsf{G}_\mu\mathbf{1} = 0, \quad M\mathsf{G}_\mu = \mathsf{G}_\mu^T M.$$

For neutral loads q, r, $q^T K_\mu r = q^T H^+ r$: the pairing is gauge independent. Changing the reference mass while holding a function f fixed changes the load Mf and the dynamical operator. It is not just a harmless recentering of the same motion.

For an injective synthesis $J : \mathbb{R}^k \to \mathbb{R}^V$, Volume 21's actual physical Gram and symmetry propositions [3] specialize to

$$G_J = J^T M J, \quad K_J = \tfrac{1}{2} J^T H J, \quad B_J = G_J^{-1} K_J, \quad G_J B_J = B_J^T G_J. \tag{3.2}$$

The orthogonal projection is $\mathcal{P}_J = J G_J^{-1} J^T M$. For every $c \in \mathbb{R}^k$ and $\lambda \in \mathbb{R}$, its residual formula is

$$\|(A - \lambda) J c\|_M^2 = \|(B_J - \lambda) c\|_{G_J}^2 + \|(I - \mathcal{P}_J) A J c\|_M^2. \tag{3.3}$$

The proof is the physical orthogonal decomposition, since $(I - \mathcal{P}_J) J c = 0$. A small compressed residual omitting the second term cannot verify the full equation. For a square change $y = Tc$, $G_y = T^{-T} G_J T^{-1}$, $K_y = T^{-T} K_J T^{-1}$, and the dual load is $T^{-T} J^T M f$, exactly as in [5], `eq:transport`.

This account gives a concrete diagnostic for a proposed clock: the pair $(M, H/2)$, not H alone, determines A. Multiplying mass by $b > 0$ while keeping stiffness fixed divides A by b. Multiplying stiffness by $c > 0$ while keeping mass fixed multiplies it by c. Multiplying both by the same scalar leaves it unchanged. With an unnormalized limiting quantum area $b\mu_h$ and the same energy, the associated motion is the unit-mass motion observed at t/b. These are algebraic time-change facts, not a theorem that a new random surface ensemble has the required limit.

The strong-not-norm example in [4], label `v21c01:strong-not-norm`, uses a moving spectral direction to show why fixed-input convergence does not control moving inputs. We use that warning in a different, positive setting: point-start transfer below requires a supremum norm for each fixed continuous resolvent test and a separate all-vertex exit estimate. It does not follow from L^2 resolvent convergence alone. We neither transplant the private indefinite pollution example to a graph Laplacian nor claim new spectral pollution here.

4 The Source Chain Before the Green Bound

4.1 Exact incidences, not approximate distances

The source's `02-inputs.tex` reconstructs primal vertices by $i \sim_L j$ when $L_i = L_j = \min_{[i,j]} L$, retains horizontal tree edges, and adds an edge for each matched vertical up/down pair. Parallel matched pairs remain distinct. Coordinate exchange gives the dual. A length-$2n$ quadrant excursion encodes the finite uniform pair; the bilateral Poissonized contour at rate m, scaled by $m^{-1/2}$, has independent limiting Brownian coordinates of variance $1/2$.

Finite instruction packages retain whole increment strings, cuts, and relative-height forest offsets. The complete-core lemma `inp:complete-core` uses selected consecutive runs, positive slack at omitted gaps, and an *exact discrete minimum* at a three-visit contact. An approximate cutoff height at that contact could admit an extra edge. Its selector lemma sandwiches spatial and value constraints with strict margins; the countable retained test library permits later selection of continuous targets on one event. These are stronger requirements than path-distance accuracy.

For the conditional-kernel transfer, the finite density is a product of gap transition masses divided by reference offset lattice masses. Changing forest coordinates is unimodular, so no lattice multiplicity is lost. Local central limits apply on compact input sets with gaps bounded below; the omitted density mass is controlled by its probability under the actual law. The weight depends only on inputs, so it does not change the output kernel. This explains the source's enlargement from

passage tests to compactified energy minima. It is not a generic claim that conditional independence survives weak limits.

Local frontier increments, incoming/outgoing tags, and seam heights must then be readable from the two fields in a buffer. Spatial product kernels, fixed quantum-unit changes, local Gaussian equivalence, and primal–dual reflection are imported from or adapted to the metric companion. We have not independently certified those field-locality and topological assertions. They are explicit boundaries of every later global claim in this report.

4.2 Nondegeneration and local energy

For a finite simple-path family $\mathcal{P}$, the source proves

$$\mathrm{EL}(\mathcal{P}) = \min_{\nu} \sum_{e} j_\nu(e)^2,$$

where j_ν is unsigned edge traffic of a probability measure on paths. The nearest point of the convex hull of incidence vectors supplies the lower bound; averaging and Cauchy–Schwarz give the reverse bound. For wired endpoints this is reciprocal condenser energy.

The nondegeneration argument assumes a sequence of vanishing annular EL quantiles b_m. The imported all-point aligned-annulus lemma yields barrier densities with squared norm $O(1/b_m)$ and enough nested shells to control projected displacement. A low-traffic flow with squared norm $O(b_m)$ would have a nonzero limiting average arclength measure Λ satisfying $\Lambda(U) \le C\sqrt{|U|}$. In particular Λ is absolutely continuous. Tangents then supply narrow-tube crossing witnesses at almost every point of a positive-area set. Point-germ triviality and the local product kernels make the witness deterministic almost everywhere; reflection and rotation produce a perpendicular dual witness. The topological crossing rule gives

$$1 \le \sum_{e} j_m(e) j_m^*(e^*) \le \|j_m\|_2 \|j_m^*\|_2 = O(b_m),$$

a contradiction. All radius choices precede the resolution limit. This is the source's argument, not a new Ram nondegeneration theorem.

Bounded ring cutoffs on the dual provide separating primal level circuits by coarea and Cauchy–Schwarz. With J disjoint rings, one such circuit has oscillation $O(\sqrt{E/J})$ for a function of energy at most E; the maximum principle propagates this inward for harmonic functions. The source thereby obtains harmonic compactness without regular face shapes.

For continuous targets it defines an inner relaxed energy by infimizing graph energy subject to uniform tolerance on compact sets, then exhausting compacts and decreasing tolerances. Countable selectors equate the liminf and limsup versions. The exact edge identity

$$d(bv + (1-b)w) = \bar{b}\, dv + (1-\bar{b})\, dw + \overline{v-w}\, db$$

gives gluing, with error bounded by a constant times $\|v-w\|_\infty^2 \mathfrak{e}(b)$. Quadraticity, local measure properties, smooth upper bounds, and an intersecting-circuit coercivity argument yield $F(u,U) = \int_U \nabla u^T A \nabla u$. The matrix A is locally elliptic. Its energy-density point germs and coordinate symmetries give $A = A^* = \lambda I$ for a deterministic positive scalar within the extraction. This identification is still conditional on the locality and nondegeneration chain just described.

4.3 Why the scalar is one

In an annulus $r < |z| < R$, let $C_0 = 2\pi/\log(R/r)$. The condenser energy tends to λC_0. A potential with additive angular period P has round energy at least P^2/C_0: integrate $(\int_0^{2\pi} \partial_\theta v \,\mathrm{d}\theta)^2 \le$

$2\pi \int_0^{2\pi} |\partial_\theta v|^2 \, \mathrm{d}\theta$ radially. Rotating the discrete condenser current gives a dual potential with period equal in magnitude to its condenser energy. Compactness and the dual lower bound therefore yield

$$\lambda C_0 \geq \lambda(\lambda C_0)^2/C_0, \quad \text{so} \quad \lambda \leq 1.$$

For the other direction, the source constructs a closed angular cochain of *exact* period one. In local branches $p_i = \theta_i/(2\pi)$, it writes $\alpha_m = a_m + d\sum_i b_{i,m}(v_{i,m} - p_i \circ \pi_m)$. The gradient correction changes no period, and recovered partition weights cancel the uncontrolled canonical cochain in the energy calculation. Its energy is at most $\lambda/C_0 + o(1)$. The signed primal–dual pairing with the condenser gradient is exactly one, so Cauchy–Schwarz gives $1 \leq \lambda^2$. Both inequalities are needed. This is a classical planar current/period argument with a source-specific recovery construction, not a consequence of isotropy alone; compare [13, 12].

4.4 The fixed-area sphere and omitted point

The companion's excursion density for an interior increment block is a continuous function of its minima and net increment in the limit, with a uniform L^2 density bound. The source retains arbitrary compactified block-measurable energy data *before* the change of density. Truncation then transfers the joint limit; no continuity of the energy-output map is asserted. Interior interval surfaces exhaust the sphere away from the tour endpoint.

At that point, continuous finite-energy functions have a removable Sobolev singularity. For recovery, flatten u near p using $u_r = u(p) + (1 - \chi_r)(u - u(p))$. Then

$$\int |\nabla(u_r - u)|^2 \leq 2 \int_{B(p,2r)} |\nabla u|^2 + C \operatorname{osc}_{B(p,2r)}(u)^2 \longrightarrow 0.$$

Local recovered cutoffs on fixed annuli permit discrete gluing with no extra energy at p. This yields the global uniform liminf/recovery and harmonic compactness statements `sph:energy`. The companion tour correspondence further supplies, in the same extraction, uniform metric distortion tending to zero, dense vertex images, weak degree-mass convergence, and vanishing edge mesh (`sph:comparison`). These imported geometric conclusions are not supplied by the finite Green calculation next.

5 The Finite-Map Green Mechanism

5.1 An independently checked finite identity

Let $\mathcal{M}$ be a connected multigraph, H its Laplacian, μ a strictly positive probability, q a zero-sum load, and $Hu = q$, $\mu(u) = 0$. For each uniform spanning tree T and $e \in T$, choose either side S_e of $T \setminus e$ and let b_e be the number of original graph edges across its cut. Then the source identity is

$$u(x) = \mathbb{E}_{T|\mathcal{M}} \sum_{e \in T} \frac{q(S_e)(\mathbf{1}_{S_e}(x) - \mu(S_e))}{b_e}. \tag{5.1}$$

The component choice changes both signed factors and hence changes no term. Parallel edges are counted separately in b_e; loops never cross a cut. The one-vertex sum is empty.

Here is a direct derivation. With D the edge-by-vertex incidence matrix, delete one vertex row of D^T, obtaining B, and delete the matching coordinate of q, obtaining $\widehat{q}$. Cauchy–Binet applied to $[B, \sqrt{z}\widehat{q}]$ shows that the coefficient of z in $\det(BB^T + z\widehat{q}\widehat{q}^T)$ is $\sum_F q(S_F)^2$, over two-component spanning forests. Leaf elimination makes the relevant determinant $\pm q(S_F)$. The determinant

lemma makes the same coefficient $\tau\widehat{q}^T(BB^T)^{-1}\widehat{q}$, where τ is the tree count. Polarization gives $q^TH^+r = \tau^{-1}\sum_F q(S_F)r(S_F)$ for neutral q, r. Each forest has exactly $b(F)$ tree extensions. Regroup the forest sum over pairs (T, e), dividing by b_e, and take $r = \delta_x - \mu$. This proves (5.1). The underlying matrix-tree method is classical [11], and the formula is already in the assigned source.

5.2 Why the tree-weighted law matters

Set $N = 2n$. The contour step-start vertex v_i visits each vertex exactly d_v times: horizontal traversals count tree incidences and matched vertical steps count non-tree incidences, including both ends of loops. A side of a tree cut occupies one cyclic interval I_e. Choose its smaller side, so $\ell_e = |I_e| \le N/2$, and then

$$\mu_n(S_e) = \ell_e/N, \qquad b_e = 1 + D_e^*. \tag{5.2}$$

The one is the cut tree edge. The other crossing edges are precisely vertical matched pairs separated by I_e, equivalently the dual-tree edges traversed an odd number of times between its two endpoints. Their number is the dual-tree distance D_e^*.

Given the number J of horizontal slots, their positions are uniform; given these positions, the horizontal and vertical strings are independent uniform Dyck excursions. With Catalan numbers C_k,

$$\#\{J = 2k\} = \binom{2n}{2k} C_k C_{n-k}, \qquad \sum_{k=0}^{n} \binom{2n}{2k} C_k C_{n-k} = C_n C_{n+1}.$$

Thus balanced counts have exponentially high probability up to a polynomial factor. This conditional independence is special to the spanning-tree encoding; it cannot be asserted for an arbitrary planar map ensemble.

For a uniform plane tree with contour length s and deterministic cyclic separation j, rerooting makes the distance D a Dyck height. Reflection gives, with parity respected,

$$p_l(a) = 2^{-l}\left[\binom{l}{(l+a)/2} - \binom{l}{(l+a+2)/2}\right], \quad \mathbb{P}(D = a) = p_j(a)p_{s-j}(a)/p_s(0).$$

Gaussian binomial bounds give $\mathbb{P}(D = a) \le C(a+1)^2 d^{-3/2} e^{-ca^2/d}$ with $d = 1 + \min(j, s-j)$, hence $\|(1+D)^{-1}\|_2 \le Cd^{-1/2}$ and $\|D\|_p \le C_p\sqrt{d}$. Dyadic chaining of fourth moments bounds the contour-segment diameter in L^4 by $C\sqrt{L}$.

For dyadic $M \le N/2$, put

$$U_{M,t} = \sum_{\substack{M \le \ell_e < 2M \\ t \in I_e}} b_e^{-1}.$$

The source's `gr:one-scale` proves $\mathbb{E}U_{M,t}^2 \le C$, uniformly in n, M, t. On a good schedule, every candidate interval and its complement contain at least cM vertical steps. Conditional on that schedule and the horizontal tree, Minkowski and the dual inverse-distance bound yield $\|U_{M,t}\|_2 \le CK_{M,t}/\sqrt{M}$. Smaller cut sides containing v_t are nested; their union cannot be the whole graph because each has mass at most $1/2$ and their intersection has positive mass. Their count $K_{M,t}$ is bounded by the horizontal-tree diameter over a deterministic $O(M)$ contour window, whose second moment is $O(M)$. Exceptional schedules have probability $O((M+1)^2 e^{-cM})$ and $U_{M,t} \le C(M+1)$ there. Unbalanced counts are controlled by $U_{M,t} \le N$ and their exponential tail. These steps close the stated one-scale bound; they do not establish the unrelated field-locality inputs.

A deterministic grid with $O(N/M)$ points controls every cut interval at that scale. Consequently the source defines

$$W_M = \sup_x \sum_{M \le \ell_e < 2M} \frac{\mathbf{1}_{S_e}(x) + \mu_n(S_e)}{b_e}, \qquad \mathbb{E}_{\text{pair}} W_M^2 \le C_0 N/M. \tag{5.3}$$

Indeed, the source's deterministic domination is $W_M \le C \max_{t \in \Gamma_M} U_{M,t}$ with $|\Gamma_M| \le CN/M$. Squaring this pointwise bound and using $\max_t U_{M,t}^2 \le \sum_t U_{M,t}^2$ proves (5.3). No independence of grid values is used, and no random maximizing time is substituted into the deterministic-time one-scale estimate. The source uses the resulting first-moment bound. It then replaces $\min(2M/N, m)$ by $(2M/N)^{1-\rho} m^\rho$, summing only for $\rho < 1/2$. We keep the minimum instead.

6 A Retained-Scale Green Modulus

The results in this section use (5.1) and (5.3) as explicit inputs. Their conclusion is a new local refinement relative to the pinned source, not a new derivation of those inputs. Let M_0 be the largest power of two not exceeding n, $\alpha_0 = 2M_0/N \in (1/2, 1]$, and $\alpha_k = \alpha_0 2^{-k}$ for $0 \le k \le K_n = \log_2 M_0$. Write $W_k = W_{M_0 2^{-k}}$ and $Z_k = \sqrt{\alpha_k} W_k$. Equation (5.3) gives $\sup_{n,k} \|Z_k\|_{L^2(\text{pair})} \le C_1 = \sqrt{2C_0}$. This is an unconditional pair-law bound, not a uniform conditional bound for every realized map.

Theorem 6.1 (Fixed-mass L^2 endpoint)**.** *For $0 \le m \le 1$ define the map-measurable functions*

$$B_n(m) = \mathbb{E}_{T|\mathcal{M}_n} \sum_{k=0}^{K_n} \min(\alpha_k, m/2) W_k, \tag{6.1}$$

$$\Phi_n(m) = \sup\{\|\mathsf{G}_n f\|_\infty : f \in \mathbb{R}^{V_n},\ \mu_n(f) = 0,\ \|f\|_\infty \le 1, \mu_n(|f|) \le m\}. \tag{6.2}$$

Then B_n is finite, nondecreasing, concave, and $B_n(0) = 0$. Simultaneously for every $m \in [0,1]$, $\Phi_n(m) \le B_n(m)$. There is a deterministic $C_2 < \infty$, independent of n, m, such that

$$\|\Phi_n(m)\|_{L^2(\text{map})} \le \|B_n(m)\|_{L^2(\text{map})} \le C_2 \sqrt{m} \quad \textit{for every deterministic } m \in [0,1]. \tag{6.3}$$

In particular $\mathbb{E}\Phi_n(m)^2 \le C_2^2 m$, $\mathbb{E}\Phi_n(m) \le C_2\sqrt{m}$, and $\mathbb{P}\{\Phi_n(m) > t\sqrt{m}\} \le C_2^2/t^2$ for deterministic $m > 0$ and $t > 0$.

Proof. For a centered one-bounded f, its positive and negative charges $\mu_n(f^+)$ and $\mu_n(f^-)$ both equal $\mu_n(|f|)/2$. Thus every subset S satisfies

$$|(M_n f)(S)| \le \min(\mu_n(S), \mu_n(|f|)/2).$$

Apply this to each cut side in (5.1), take absolute values, and use $\mu_n(S_e) = \ell_e/N < \alpha_k$ on scale k. Maximizing over x gives the simultaneous bound by (6.1). Concavity and monotonicity follow termwise. The supremum defining Φ_n is over a compact finite-dimensional polytope and a continuous objective; measurability is immediate, also because there are finitely many map states at fixed n. The finite conditional tree averages define all masses on one version, with no uncountable exceptional-set issue.

For deterministic $h = m/2 > 0$, put $c_k(h) = \min(\alpha_k, h)/\sqrt{\alpha_k}$. Conditional expectation is an L^2 contraction, so Minkowski's inequality for the finite sum gives

$$\|B_n(m)\|_{L^2(\text{map})} \le \|\sum_{k=0}^{K_n} c_k(h) Z_k\|_{L^2(\text{pair})} \le C_1 \sum_{k=0}^{K_n} c_k(h).$$

No cross terms are discarded and no independence is needed. Extend only the deterministic comparison scales to all $k \geq 0$ and choose the last $j \geq 0$ with $\alpha_j \geq h$. This exists since $h \leq 1/2 < \alpha_0$. The scales above h contribute

$$\sum_{k=0}^{j} \frac{h}{\sqrt{\alpha_k}} \leq \frac{\sqrt{h}}{1 - 2^{-1/2}};$$

the finer ones contribute $\sum_{k>j} \sqrt{\alpha_k} \leq \sqrt{h}/(1 - 2^{-1/2})$. This proves (6.3), for example with $C_2 = \sqrt{2}C_1/(1 - 2^{-1/2})$. The case $m = 0$ is zero, and Markov's inequality applied to the square proves the probability statement; Cauchy–Schwarz gives the first moment. This proof remains valid for $j > K_n$, without introducing any additional random scale. □

Theorem 6.2 (One coefficient for all sources)**.** *For every fixed real $p > 1$, define*

$$a_{n,p} = \left[\mathbb{E}_{T|\mathcal{M}_n} \sum_{k=0}^{K_n} \frac{Z_k^2}{(k+1)^p} \right]^{1/2}. \tag{6.4}$$

Then $\sup_n \mathbb{E}a_{n,p}^2 \leq C_1^2 \zeta(p)$, so $(a_{n,p})_n$ is tight and uniformly integrable in L^1. There is a deterministic constant $C_p < \infty$ such that, simultaneously for all centered real one-bounded f on each map,

$$\|\mathsf{G}_n f\|_\infty \leq C_p a_{n,p} \sqrt{\mu_n(|f|)}[1 + \log(1/\mu_n(|f|))]^{p/2}. \tag{6.5}$$

The right side is defined as zero when $\mu_n(|f|) = 0$. The same bound, with mass m, holds for $B_n(m)$ for every $m > 0$.

Proof. The second-moment bound follows from the tower property, $\mathbb{E}Z_k^2 \leq C_1^2$, and the summability of $(k+1)^{-p}$. Moreover

$$\sup_n \mathbb{E}[a_{n,p} \mathbf{1}_{\{a_{n,p} > R\}}] \leq C_1^2 \zeta(p)/R \longrightarrow 0.$$

This proves the asserted L^1 uniform integrability, not uniform integrability of the squares. Conditional Cauchy–Schwarz gives

$$\mathbb{E}[Z_k \mid \mathcal{M}_n] \leq \left(\mathbb{E}[Z_k^2 \mid \mathcal{M}_n]\right)^{1/2} \leq (k+1)^{p/2} a_{n,p}.$$

It suffices to bound

$$S_p(h) = \sum_{k \geq 0} \frac{\min(\alpha_k, h)}{\sqrt{\alpha_k}} (k+1)^{p/2}, \qquad h = m/2.$$

With j as above, $\alpha_j \in [h, 2h)$ and $j + 1 \leq C[1 + \log(1/m)]$. For $k \leq j$, the summand is at most $\sqrt{h}\, 2^{-(j-k)/2}(j+1)^{p/2}$. For $k = j + r$, $r \geq 1$, it is at most $\sqrt{2h}\, 2^{-r/2}(j+1)^{p/2}(r+1)^{p/2}$. The two geometric sums are finite, the second because exponential decay dominates any fixed power. This proves $S_p(h) \leq C_p \sqrt{m}[1 + \log(1/m)]^{p/2}$. The finite source scale set is bounded by this infinite sum even when m is smaller than every discrete scale. Applying the pointwise envelope from Theorem 6.1 completes the proof. Equivalently, writing $d_k = \mathbb{E}[Z_k \mid \mathcal{M}_n]$, weighted Cauchy–Schwarz gives

$$B_n(m) \leq a_{n,p} \left[\sum_{k=0}^{K_n} c_k(h)^2 (k+1)^p \right]^{1/2},$$

because $\sum_k d_k^2/(k+1)^p \leq a_{n,p}^2$. The same split bounds the bracket by $C_p m (j+1)^p$. All conditional averages here are on the same full realized map. □

What the improvement means. For every fixed $\rho < 1/2$ and $p > 1$, $m^{1/2}[1 + \log(1/m)]^{p/2} = o(m^{\rho})$ as $m \downarrow 0$. Thus (6.5) improves the mass dependence obtainable from the source's same one-scale second-moment input. Its coefficient is different; we do not claim it is pointwise smaller than the source coefficient. Every fixed logarithmic exponent $\theta = p/2 > 1/2$ is available; constants are not uniform as $\theta \downarrow 1/2$. No claim of the exact exponent $1/2$ or of optimality is made. The coefficient has uniformly integrable first powers, but UI of its squares is not proved. Equation (6.3) at deterministic m cannot be substituted at a map-dependent random m without further control; the simultaneous bound (6.5) is the appropriate statement in that situation. Neither result proves a common tight coefficient for a pure $\sqrt{m}$ bound at every m. The concave modulus is B_n; the displayed logarithmic majorant need not be nondecreasing on $(0, 1]$.

A concrete use: sparse residuals. If $H_n u = M_n f$, $\mu_n(u) = 0$, and a centered approximation $\widehat{u}$ has residual $r = M_n^{-1} H_n \widehat{u} - f$, then $\mu_n(r) = 0$ and $\widehat{u} - u = \mathsf{G}_n r$. For $F = \|r\|_\infty > 0$,

$$\|\widehat{u} - u\|_\infty \le F B_n(\mu_n(|r|)/F). \tag{6.6}$$

If r is supported on a set of mass at most deterministic s, this is at most $F B_n(s)$. For deterministic F, or a deterministic upper bound on it, the L^2 error over maps is at most $C_2 F\sqrt{s}$, and the expected error obeys the same bound. For map-dependent residual amplitudes no product-of-expectations claim is made. The deterministic inequality and (6.5) remain valid. This is a direct consequence of the same modulus, not a separately counted contribution or an efficient solver.

7 The Source's Uniform Operator Bridge

Fix a deterministic sequence of connected finite graphs, positive probabilities μ_n, and maps $\pi_n : V_n \to S^2$. This paragraph states the source's hypotheses precisely, with μ a nonatomic full-support probability on the sphere.

H1. $(\pi_n)_* \mu_n \Rightarrow \mu$ and the maximum projected edge length in round distance tends to zero.

H2. Every sequence uniformly converging under π_n to continuous u has $\liminf \mathfrak{e}_n(u_n) \ge \int |\nabla u|^2$, with infinity unless $u \in W^{1,2}(S^2)$. Every smooth u has uniform recovery with energy tending to this integral.

H3. On every open set, graph-harmonic sequences with uniformly bounded supremum norm and total energy are locally equicontinuous and locally uniformly precompact, with comparison at every vertex over each compact subset, not merely almost every vertex.

H4. For some $C < \infty$ and $\rho > 0$ independent of n, $\|\mathsf{G}_n g\|_\infty \le C \mu_n(|g|)^\rho$ for all centered one-bounded g. The continuum half-energy form has the closure, regularity, conservativity, and continuous-path hypotheses stated at (2.4).

The chain here has generator $\mathcal{L}_n = -M_n^{-1} H_n/2$. The source's deterministic theorem `dyn:conditional` proves uniform convergence of $R_\alpha^n = (\alpha - \mathcal{L}_n)^{-1}$ on continuous data, for every fixed $\alpha > 0$, and stationary path convergence. We reconstruct the parts needed for the extension.

For $u_n = R_\alpha^n(f \circ \pi_n)$, the maximum principle and the resolvent equation give

$$\|u_n\|_\infty \le \|f\|_\infty/\alpha, \quad H_n u_n = M_n g_n, \quad g_n = 2(f \circ \pi_n - \alpha u_n).$$

Both energy and $\|g_n\|_\infty$ are bounded. Near a point choose disjoint continuity neighborhoods U, V with $\mu(V) > 0$, making $\mu(U)$ small, and remove the centered source

$$q_n = g_n \mathbf{1}_{U_n} - \frac{\mu_n(g_n \mathbf{1}_{U_n})}{\mu_n(V_n)} \mathbf{1}_{V_n}, \qquad U_n = \pi_n^{-1} U,\ V_n = \pi_n^{-1} V.$$

Its supremum is bounded and $\mu_n(|q_n|) \le C\mu_n(U_n)$. H4 makes $\mathsf{G}_n q_n$ uniformly small; its energy is bounded by $\|\mathsf{G}_n q_n\|_\infty \mu_n(|q_n|)$. The difference $u_n - \mathsf{G}_n q_n$ is harmonic on U_n and satisfies H3. Choosing the mass before the equicontinuity scale yields global uniform compactness. Full support and H1 guarantee dense vertex images.

A subsequential continuous limit minimizes

$$J(u) = \mathcal{E}(u,u) + \alpha\mu(u^2) - 2\mu(fu).$$

Indeed the discrete minimizer has functional $\mathfrak{e}_n(u)/2 + \alpha\mu_n(u^2) - 2\mu_n((f \circ \pi_n)u)$, and H2 supplies the lower bound and smooth competitors. Smooth functions are a form core. Strict convexity identifies the unique $L^2(\mu)$ minimizer $R_\alpha f$, and full support identifies its unique continuous version. Consequently

$$\|R_\alpha^n(f \circ \pi_n) - (R_\alpha f) \circ \pi_n\|_\infty \to 0. \tag{7.1}$$

The contraction bound $\|R_\alpha^n\|_\infty \le 1/\alpha$ gives the same result for right-hand sides uniformly converging to f.

The source also proves

$$\|\alpha R_\alpha f - f\|_\infty \to 0 \quad (\alpha \to \infty), \qquad f \in C(S^2). \tag{7.2}$$

There is a real measure issue here: convergence in the μ form norm does not directly imply round L^2 convergence. For smooth f, the spectral theorem gives gradient convergence. Round Poincare leaves only a bounded additive constant. The Markov range bounds $\min f \le \alpha R_\alpha f \le \max f$ force each subsequential constant to zero, by testing the essential round extrema of f. Round Sobolev convergence then supplies a surrounding circle with small one-dimensional H^1 trace error. Testing the weak resolvent equation with positive and negative parts on its interior bounds the interior values by boundary values and the local extrema of f. Small disks and a finite cover prove (7.2); uniform smooth approximation extends it to all continuous f.

With finitely many cutoff functions separating points at round distance $d > 0$, (7.1)–(7.2) give $v_n = \alpha R_\alpha^n(f \circ \pi_n)$ uniformly within $b > 0$ of the cutoff, and $\|\mathcal{L}_n v_n\|_\infty \le C_b$. Stopping its martingale at the first displacement d or time s gives, *uniformly over every starting vertex*,

$$\sup_v \mathbb{P}_v\{\tau_d^n(v) \le s\} \le 2b + C_b s \quad \text{for all sufficiently large } n. \tag{7.3}$$

First $n \to \infty$, then $s \downarrow 0$, then $b \downarrow 0$. Strong Markov and Aldous's criterion [14] give tightness for *any* sequence of starts, not just stationary starts. Small projected jumps make all J_1 limits continuous. The projection itself need not be Markov.

Proposition 7.1 (A modulus suffices)**.** *The above argument remains valid if the power bound in H4 is replaced by $\|\mathsf{G}_n g\|_\infty \le C\omega(\mu_n(|g|))$, for all centered one-bounded g, where ω is bounded on $[0,1]$ and tends to zero at zero. No monotonicity beyond a nondecreasing majorant is needed.*

Proof. Scaling a source of norm at most F bounds its potential by $CF \sup_{t \le \mu_n(|q_n|)/F} \omega(t)$. This tends to zero when the neighborhood mass does. Every other compactness, variational, and exit step just given is unchanged. The nondecreasing majorant tends to zero as well and is bounded. □

In the source's random extraction, tightness of $a_{n,p}$ permits a further Skorokhod coupling where it converges to a finite value and is bounded. Theorem 6.2 then supplies this deterministic modulus hypothesis on the coupled realization. This uses tightness, not an unjustified almost-sure bound in the original probability space, and does not remove any of H1–H3 or the continuum form assumptions.

8 Conditional Transfer to Every Starting Point

We now use only the conclusions (7.1), (7.3), dense vertex images, and vanishing projected jumps. This separation makes clear which extra input beyond a stationary theorem is needed. Write $Q^T_{n,v}$ for the projected path law from vertex v and let $d_{\mathrm{BL},T}$ be a bounded-Lipschitz metric for weak convergence on $D([0,T],S^2)$ with a bounded compatible J_1 metric.

Theorem 8.1 (All-start conditional extension)**.** *Under H1–H4, for every finite $T>0$ there is a unique weakly continuous family $x\mapsto Q^T_x$, $x\in S^2$, of continuous-path probability laws starting at x, characterized by the nested-resolvent identities* (8.1) *below. The families are consistent in T. They have the following properties.*

(i) *If $\pi_n(v_n)\to x$, then $Q^T_{n,v_n}\Rightarrow Q^T_x$.*

(ii) $\sup_{v\in V_n} d_{\mathrm{BL},T}(Q^T_{n,v},Q^T_{\pi_n(v)})\to 0$.

(iii) *For arbitrary probability measures ν_n on V_n with $(\pi_n)_*\nu_n\Rightarrow\nu$, the projected path law from ν_n converges to $\int Q^T_x\nu(\,\mathrm{d}x)$.*

The consistent family is the canonical Feller continuous-path version of the source diffusion: it agrees with its conditional starting laws for μ-almost every x, and extends them continuously to every x. No claim about an arbitrarily altered exceptional-set version of an L^2-defined process is intended.

Proof. Existence and uniqueness at a point. For each x, choose v_n with $\pi_n(v_n)\to x$, using density. The uniform exit condition and small jumps give subsequential continuous-path limits, diagonally on all positive integer horizons. Every such law starts at x. For continuous $f_0,\dots,f_k$ and positive $\beta_1,\dots,\beta_k$, the finite chain Markov property gives its integrated finite-dimensional expectation as

$$(f_0\circ\pi_n)(v_n)R^n_{\beta_1}\big((f_1\circ\pi_n)R^n_{\beta_2}(\cdots R^n_{\beta_k}(f_k\circ\pi_n)\cdots)\big)(v_n).$$

Successive use of (7.1), including varying data, makes this converge to

$$f_0(x)R_{\beta_1}\big(f_1R_{\beta_2}(\cdots f_{k-1}R_{\beta_k}f_k\cdots)\big)(x). \tag{8.1}$$

Specifically, this is the integral of $\mathbb{E}_Q\prod_{i=0}^k f_i(X_{t_1+\cdots+t_i})$ against $\prod_{j=1}^k e^{-\beta_j t_j}\,\mathrm{d}t_j$, with the empty sum zero. Weak path convergence justifies integration on bounded time boxes; bounded test functions and exponential tails justify removing those boxes. Product expectations of continuous-path laws are continuous in all gaps, including zero. Uniqueness of multivariate Laplace transforms, applied one coordinate at a time, equates them everywhere. Product tests determine finite-dimensional laws, which determine the law on continuous paths. Thus all subsequential laws coincide, independently of the selected vertices. This proves (i) and consistency.

Continuity of the family. Suppose $x_j\to x$. For each fixed j, part (i) permits a sufficiently large index $n_j>n_{j-1}$ and a vertex v_j with $d_0(\pi_{n_j}(v_j),x_j)<1/j$ and $d_{\mathrm{BL},T}(Q^T_{n_j,v_j},Q^T_{x_j})<1/j$. The sequence $\pi_{n_j}(v_j)$ tends to x, so (i), which applies to every subsequence, makes $Q^T_{n_j,v_j}\Rightarrow Q^T_x$. The triangle inequality proves $Q^T_{x_j}\Rightarrow Q^T_x$.

Uniformity and mixtures. If (ii) failed, choose offending v_n along a subsequence. Compactness gives a further subsequence with $\pi_n(v_n) \to x$. Part (i) and the just proved continuity make both compared laws tend to Q_x^T, a contradiction. For a bounded Lipschitz path functional F, (ii) controls the difference of its expectations under $Q_{n,v}^T$ and $Q_{\pi_n(v)}^T$ uniformly in v. The function $x \mapsto \int F \, \mathrm{d}Q_x^T$ is continuous. Integrating and using weak convergence of $(\pi_n)_*\nu_n$ proves (iii). Bounded Lipschitz tests determine weak convergence on this Polish path space.

Identification and the Feller property. Let $P_t f(x) = \int f(X_t) \, \mathrm{d}Q_x$ for continuous f. Evaluation at t is continuous at the continuous paths supporting these laws, so $P_t f$ is continuous. Part (i) and compactness imply

$$\sup_v |P_t^n(f \circ \pi_n)(v) - (P_t f)(\pi_n(v))| \to 0.$$

The discrete Markov identities pass to the limit using this uniform convergence and the contraction of P_s^n. They give $P_{s+t} = P_s P_t$ and the corresponding finite-dimensional Markov identities for Q_x. Positivity and $P_t 1 = 1$ pass as well. Strong continuity at zero is uniform on x: for a continuous f, separate its modulus of continuity at distance d from $2\|f\|_\infty$ times the probability of an exit by t, and use (7.3). Passage to the limit may use the open event of displacement greater than d, then a smaller discrete threshold. Hence $\|P_t f - f\|_\infty \to 0$.

Finally (iii) with $\nu_n = \mu_n$ and the source's stationary identification give $\int Q_x \mu(\mathrm{d}x) = \mathrm{Law}_\mu(B)$. Disintegrating with respect to the initial coordinate proves agreement for μ-almost every x. Equivalently the Laplace resolvent of P_t is R_α, which identifies its L^2 semigroup. Full support and continuity make this extension unique. An associated regular Dirichlet-form process is initially specified outside an exceptional set; the continuous family just constructed is the version meant here. □

Corollary 8.2 (Application on a coupled quantum sphere)**.** *Grant the source's common metric/energy extraction, Green one-scale estimate, and continuum-form assumptions. On each resulting probability-one coupled realization satisfying H1–H4, the conclusion of Theorem 8.1 holds, in round or D_h coordinates. In particular it applies to map-dependent chosen vertices whose images converge, and to arbitrary weakly convergent initial probabilities on that realization. The same acceleration is n.*

Proof. The Green constants may be retained during tight extraction as in Section 7. Round and D_h topologies agree on a compact sphere, hence their identity maps are uniformly continuous. The common metric correspondence has uniformly vanishing vertex displacement. Applying it at every path time gives vanishing J_1 discrepancy with the identity time change. Neither changing the initial law nor this metric transport changes (2.2). □

For random environments, an annealed statement for specially chosen starts additionally requires a joint convergence specification of those starts with the environment. The corollary does not invent one. It also does not assert convergence for a vertex sequence whose projected positions have no limit; its subsequential limits have the corresponding subsequential starting points.

Unlike an initial-density tilt of stationary laws, part (iii) allows $\nu_n = \delta_{v_n}$ even when $1/\mu_n(v_n)$ diverges. The reason is the *uniform* Green/resolvent/exit information, not a new change-of-measure trick. A stationary theorem alone cannot supply it: the reversible two-state chain with rates $q(0,1) = 1/n$, $q(1,0) = n$ has stationary mass $1/(n^2+1)$ at 1 and converges stationarily to the constant zero path, but starts at 1 are not tight at time zero. This inherited boundary example from M375 is only a contrast; it is not a counterexample within the source's stronger hypotheses and is not counted as a new contribution.

9 Exact Finite Verification

The accompanying `evidence/M377-parent-tests.py` uses exact Python `Fraction` arithmetic. It enumerates every spanning tree on three fixed connected multigraphs with respectively 2, 5, and 13 spanning trees, counting edge multiplicity. The graphs have 2, 3, and 4 vertices and include loops and unequal degree masses. For respectively 6, 24, and 78 normalized centered test vectors, 108 in total, it compares the forest inverse with a grounded Laplacian solve followed by mass recentering. All comparisons pass. It also checks the centered-charge $m/2$ bound, the scale-adaptive B_n envelope, and the accelerated inverse factor $2\mathsf{G}_n$ exactly. These are bounded finite tests, not a numerical proof for arbitrary graphs or all source functions. The algebraic derivations in Sections 3–6 establish the universal finite identities in their stated domains.

One nonuniform-mass example is the four-vertex cycle with a diagonal 0–2, a second 0–1 edge, and a loop at vertex 3. There are seven edges, degrees $(4, 3, 3, 4)$, and

$$M = \frac{1}{14}\operatorname{diag}(4, 3, 3, 4), \qquad H = \begin{pmatrix} 4 & -2 & -1 & -1 \\ -2 & 3 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}.$$

The loop changes the mass but not the displayed last diagonal of H. The unequal ratio $H_{33}/d_3 = 1/2$ records invisible attempts. The neutral-load forest check averages *all* spanning trees; replacing this average by just the originally selected tree would not give the map Green operator in general.

The suite does not test the Catalan schedule law, Dyck estimates, the probabilistic one-scale lemma, its universal constants, or expectations over all finite random maps. An independent narrow proof review accepts the exact envelope and the strengthened fixed-mass L^2 and simultaneous logarithmic refinements conditional on the one-scale estimate; that review is not a certification of the geometric source chain. The infinite logarithmically weighted sum in Theorem 6.2 is justified analytically by exponential summability, not by truncating a numerical experiment. No simulation is offered as evidence for GHP convergence, field-locality kernels, unit conductivity, tightness of quantum-map paths, or a universal convergence rate. Script hashes, test-vector counts, and the list of tested graph sizes are in the accompanying JSON record.

10 Current Primary Literature and Comparison

Public primary sources were checked on October 8, 2026, using only public titles or identifiers in web queries. Private corpus content was not uploaded. This is a bounded literature comparison, not an exhaustive priority determination or a claim that no related result exists.

Berestycki–Gwynne [7], Theorem 1.2 and Remark 1.3, prove quenched, time-parametrized convergence on mated-CRT maps, uniformly over compact sets of starting points in their whole-plane setting. Their deterministic clock is an annealed exit-time median, comparable to inverse cell area; the remark leaves convergence of its ratio to inverse area open. Their model and clock are different from the finite combinatorial tree-weighted sphere. Thus our conditional all-start consequence is not the first all-start LBM approximation in the literature. Its role is to expose what this pinned source's stronger uniform conclusions already permit.

Garban–Rhodes–Vargas [8], Theorem 1.7 and Remark 1.8, identify the Liouville form and explain the importance of its reference measure. Their displayed form is unhalved, so its formula must not be copied without comparing clock conventions. Their construction and discussion distinguish an all-point process from an abstract form construction on a possibly exceptional starting set; this is

why Theorem 8.1 specifies a canonical continuous version. Their intrinsic form pseudometric does not identify the desired quantum metric.

Bertacco–Gwynne–Sheffield [9], published in 2025, analyze Smith embeddings through primal and dual walks converging modulo time change. This is relevant context for electrical embedding and duality, but modulo-time convergence does not calibrate the half-edge clock here. Ooi [10], version 4 of October 2024, studies convergence of processes time-changed by Gaussian multiplicative chaos under additional Green-function hypotheses in the L^2 regime. That framework concerns a different approximation and does not replace the finite-map source estimate or certify the endpoint $\sqrt{2}$ ensemble at issue here.

The classical attributions are substantive: matrix-tree and forest expansions [11], planar electrical duality [12], self-dual conductivity ideas [13], stopping-time tightness [14], and regular Dirichlet forms [15] supply the methods. Weighted summation, conditional expectation, Laplace uniqueness, and compactness arguments are also classical. The realized additions concern the precise source-dependent modulus and transfer theorem, not ownership of these tools.

11 Advantages, Limits, and Further Work

The retained-scale calculation improves the mass dependence without requiring a stronger one-scale moment estimate. Its L^2 endpoint bound is uniform in map size at each deterministic mass, and its summable-weight version supplies a single L^2-bounded random coefficient for every admissible source. Concavity of B_n preserves a useful scale-adaptive finite modulus before it is replaced by a simpler formula. The sparse-residual interpretation ties the estimate to the actual mass and Green domain rather than coefficient magnitude.

The all-start theorem makes efficient use of a strong source conclusion: uniform exits and continuous resolvents allow moving point starts and arbitrary weak initial distributions, without a density assumption. It retains the conditional environment, the same spatial correspondence, and the exact n clock. The proof supplies a continuous all-point version instead of evaluating an unspecified L^2 representative.

There are material limitations.

- The fixed-mass L^2 square-root bound is not a simultaneous pure square-root random bound. Removing the logarithmic envelope would require additional control across scales; it is not implied by a uniform second moment for each Z_k alone.
- The improved Green modulus does not give a rate of uniform resolvent or path convergence. Harmonic compactness, recovery, projected mesh, and metric distortion are still qualitative.
- The conditional path extension does not reprove exact local incidences, Gaussian field kernels, the aligned-annulus theorem, the scalar conductivity result, or the metric companion's global limit. A failure of an imported input would invalidate an unconditional application while leaving the stated deterministic implications intact.
- The bound $\mathbb{E}a_{n,p}^2 \le C_p$ gives tightness and uniform integrability in L^1, but does not establish uniform integrability of $a_{n,p}^2$ or a useful almost-sure rate along the original sequence of map samples. Fixed deterministic-mass estimates cannot be substituted at correlated random masses inside expectations.
- Computing B_n by enumerating all spanning trees can be costly. No algorithmic speedup, conditioning advantage, or numerical complexity improvement has been measured. The Gram matrix can itself be poorly conditioned, and a reduced bank can omit a large residual.

Further work could establish stronger tail or joint-scale estimates for the forest traffic to approach a simultaneous critical modulus; quantify harmonic compactness and recovery to obtain actual operator rates; or specify measurable finite-map starting rules with a proved joint limit in the environment. These are separate tasks, not results counted here. Growing products of walks likewise need quantitative error control not supplied by convergence of each fixed product.

The realized assessment is $C3/5$, distinct from relevance $L4/5$. The sharper near-endpoint mass dependence and extension from stationary to arbitrary weak starting laws are notable local improvements under the source's declared inputs. This rating does not award credit for the imported one-scale estimate, global geometry, or conductivity theorem. The methods are classical; neither global quantum-map certification nor historical novelty is claimed.

A Read Scope and Reproducibility

The provenance record distinguishes whole-file byte hashing from actual read scope. All assigned source mathematical text was read: `paper.tex`, `bibliography.tex`, history/introduction, and sections `02-inputs` through `07-dynamics`, including their proofs. The drawings' TikZ source was read; no claim of a separate rendered inspection of the original source PDF is made.

Selected companion scope is as follows, using physical file lines.

- `tree/background.tex`, lines 43–147: Mullin encoding, ordinary fixed-area sphere convention, and peanosphere/interval surface statements with their cited dependencies.
- `tree/certificates.tex`, lines 412–478: finite-band reconstruction statement and the beginning of its exact gap argument, not its entire proof or the full conditional-kernel chain.
- `common-annuli.tex`, lines 278–370: the all-point absolute-scale annulus statement and initial proof, not certification of its Gaussian or conditional-independence inputs.
- `tree/sphere.tex`, lines 1–246 and 465 through the end: excursion densities with their proof, contour-mass identity, sphere identification proof, and deterministic normalization. The intervening endpoint-modulus proof was not read in full here and remains imported.
- Private `ch01d-physical-metrics.tex` and `ch01f-spectral-limits.tex`: full actual mathematical text; their separate plotted image files were not inspected.
- Private `TR-meta5-math.tex`: lines 115–196 and 451–524, covering graph energy, neutral loads, general gauge, transported forms/loads, exact static elimination, and the start of Laplace moments.

These are selected files found by actual local search, not a claim to have read the entire private corpus. The numerical and algebraic tests have their own bounded scope. Build records state whether native editor/compiler tools were available, identify the physical exporter, and preserve log, rendered-page, visual, and bounds checks. The final manifest uses relative nested paths and SHA-256 hashes; it excludes mutable coordination status and shared project catalogues.

References

[1] OpenAI, *A Linear Clock for Random Walk on Tree-Weighted Planar Maps*, October 5, 2026, pinned source. Public source directory.

[2] OpenAI, *Metric-measure limits of subcritical FK and spanning-tree planar maps*, September 24, 2026, same pin, selected passages specified in Appendix A. Public companion directory.

[3] M. Subbarao, *Ram Calculus*, Volume 21, `ch01d-physical-metrics.tex`, private local source. Labels `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, and `v21c01:residual-decomposition`; actual proofs read.

[4] M. Subbarao, *Ram Calculus*, Volume 21, `ch01f-spectral-limits.tex`, private local source. Label `v21c01:strong-not-norm` and subsection "Domain-aware approximate inversion"; actual example and proofs read.

[5] M. Subbarao, *Ram Green Operators and Sparse Tree Moments for Local Heights on Semistable Curves*, private `TR-meta5-math.tex`; equations `eq:gauge`, `eq:gaugeinv`, `eq:transport`, `eq:elimpair`.

[6] M. Subbarao, *Ram Physical Coordinates and Stable Extensions for Critical FK–Ising Walk Limits*, technical report TR-Ram-OpenAI-722-M375, October 8, 2026, frozen private project report.

[7] N. Berestycki and E. Gwynne, *Random walks on mated-CRT planar maps and Liouville Brownian motion*, Communications in Mathematical Physics 395 (2022), 773–857. arXiv:2003.10320; DOI.

[8] C. Garban, R. Rhodes, and V. Vargas, *On the heat kernel and the Dirichlet form of Liouville Brownian motion*, Electronic Journal of Probability 19 (2014), no. 96, 1–25. arXiv:1302.6050; DOI.

[9] F. Bertacco, E. Gwynne, and S. Sheffield, *Scaling limits of planar maps under the Smith embedding*, Annals of Probability 53 (2025), no. 3, 1138–1196. arXiv:2306.02988; DOI.

[10] T. Ooi, *Convergence of processes time-changed by Gaussian multiplicative chaos*, version 4, October 1, 2024. arXiv:2305.00734v4.

[11] S. Chaiken, *A combinatorial proof of the all minors matrix tree theorem*, SIAM Journal on Algebraic Discrete Methods 3 (1982), 319–329. DOI.

[12] R. Lyons and Y. Peres, *Probability on Trees and Networks*, Cambridge University Press, 2016, Chapters 4 and 9. DOI.

[13] A. M. Dykhne, *Conductivity of a two-dimensional two-phase system*, Soviet Physics JETP 32 (1971), 63–65. Primary article scan.

[14] D. Aldous, *Stopping times and tightness*, Annals of Probability 6 (1978), 335–340. DOI.

[15] M. Fukushima, Y. Oshima, and M. Takeda, *Dirichlet Forms and Symmetric Markov Processes*, second edition, De Gruyter, 2011. DOI.